MATTEO IOVINO, PH.D.

Post-Doctoral Researcher at ETH Zürich

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EXPERIENCE

Post-Doctoral Researcher
ETH Zürich D-MAVT
Sep 2025 – Present | Zürich, Switzerland

- Academic Guest at the Mobile Robotics Lab (MRL) led by Prof. Stefan Leutenegger
- Recipient of the WASP International Postdoctoral Scholarship.

Scientist
ABB Research
Jun 2023 – Aug 2025 | Västerås, Sweden

- Project manager of the WASP Research Arena for Robotics, interacting with stakeholders from both academia and industry
- Enhancing collaborative robots with AI-ready skills
- Lab host: resource scheduling and responsible for the safety requirements of the setups

Industrial Ph.D.
ABB Research
Feb 2019 – Jun 2023 | Västerås, Sweden

- Automatic and intuitive generation of task level policies for robotic applications

Engineering Internship
ABB Research
Apr 2018 – Sep 2018 | Västerås, Sweden

- Simulating a mobile manipulation task in ROS/Gazebo

Engineering Internship
Safran Nacelles
Mar 2017 – Aug 2017 | Le Havre, France

- Set up of a simulation environment in AMESim for a thrust reverser test bench

EDUCATION

Ph.D. in Computer Science
KTH - Royal Institute of Technology
Mar 2019 – Jun 2023 | Stockholm, Sweden

- Thesis title: *Learning Behavior Trees for Collaborative Robotics*
- Collaboration with *ABB Research*

Research Visit
ETH Zürich D-MAVT
Apr 2022 – Sep 2022 | Zürich, Switzerland

- Study period at the Autonomous Research Lab (ASL) led by Prof. Roland Siegwart

M.Sc. in Engineering
École Centrale de Nantes
2016 – 2018 | Nantes, France

- Major in Robotics. GPA: 3.82

SKILLS

Programming
- Python ●●●●○
- C++ ●●○○○

Operating Systems
- Linux ●●●●○
- MacOS ●●●●○
- Windows ●●○○○

Software & Tools
- Git ●●●●○
- ROS/ROS2 ●●●●○
- LaTeX ●●●●○

LANGUAGES

- Italian ●●●●●
- English ●●●●○
- Swedish ●●●○○
- Spanish ●●●●○
- French ●●●○○
- German ●○○○○

PROJECTS

WASP-CBSS-BT
Setup of a tutorial about Behavior Trees in Robotics for the WASP Summer School

WASP - Manipulation and Mobility
Vision-based manipulation and mobility based on Behavior Trees and Linear Temporal Logic planning

Dual Arm Manipulation
Cooperative object carrying between a mobile manipulator and a human operator

M.Sc. in Control Engineering | 110/110 cum Laude
Università degli Studi di Padova
2015 – 2018 Padova, Italy

B.Sc. in Information Engineering | 103/110
Università degli Studi di Padova
2012 – 2015 Padova, Italy

SUPERVISION EXPERIENCE

All supervision experience pertains to an industrial context.

- 2020 - M. Edin, M.Sc. Thesis (Uppsala University), Co-Supervisor
- 2021 - O. Gustavsson, M.Sc. Thesis (KTH), Co-Supervisor
- 2022 - Z. Xie, M.Sc. Thesis (KTH), Co-Supervisor
- 2024 - R. Mohan, M.Sc. Thesis (Blekinge Institute of Technology), Main Supervisor
- 2024 - N. Turcato, Ph.D. research visit (University of Padova), Main Supervisor
- 2025 - A. Adami, Industrial Ph.D. (ABB and University of Padova), Co-Supervisor

PUBLICATIONS

Journal Articles

- J. Styrud, **M. Iovino**, R. Stower, *et al.*, "Design and evaluation of an assisted programming interface for behavior trees in robotics," *Under revision.*, 2026.
- A. Adami, A. Synodinos, **M. Iovino**, R. Carli, and P. Falco, "Learning stack-of-tasks management for redundant robots," *Under revision.*, 2025.
- **M. Iovino**, J. Förster, P. Falco, J. J. Chung, R. Siegwart, and C. Smith, "Comparison between behavior trees and finite state machines," *IEEE Transactions on Automation Science and Engineering*, 2025.
- N. Turcato, **M. Iovino**, A. Synodinos, A. Dalla Libera, R. Carli, and P. Falco, "Towards autonomous reinforcement learning for real-world robotic manipulation with large language models," *IEEE Robotics and Automation Letters*, pp. 1–8, 2025. DOI: 10.1109/LRA.2025.3589162.
- **M. Iovino**, E. Scukins, J. Styrud, P. Ögren, and C. Smith, "A survey of Behavior Trees in robotics and AI," *Robotics and Autonomous Systems*, vol. 154, p. 104 096, Aug. 2022, ISSN: 0921-8890. DOI: 10.1016/j.robot.2022.104096.

Conference Proceedings

- J. Borgstedt, J. Bhattacharyya, **M. Iovino**, F. E. Pollick, and S. Brewster, "Design and integration of thermal and vibrotactile feedback for lifelike touch in social robots," in *Under revision.*, 2025.
- D. C. Domínguez, M. Iannotta, A. Kashyap, *et al.*, "The first wara robotics mobile manipulation challenge – lessons learned," in *2025 European Conference on Mobile Robotics (ECMR)*, Sep. 2025.
- J. Styrud, **M. Iovino**, M. Norrlöf, M. Björkman, and C. Smith, "Automatic behavior tree expansion with llms for robotic manipulation," in *2025 International Conference on Robotics and Automation (ICRA)*, May 2025.
- M. Hallen, **M. Iovino**, S. Sander-Tavallaey, and C. Smith, "Behavior trees in industrial applications: A case study in underground explosive charging," in *2024 IEEE 19th International Conference on Automation Science and Engineering (CASE)*, Aug. 2024.
- **M. Iovino**, J. Förster, P. Falco, J. J. Chung, R. Siegwart, and C. Smith, "On the programming effort required to generate Behavior Trees and Finite State Machines for robotic applications," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, May 2023.
- **M. Iovino**, J. Styrud, P. Falco, and C. Smith, "A framework for learning behavior trees in collaborative robotic applications," in *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)*, Aug. 2023, pp. 1–8. DOI: 10.1109/CASE56687.2023.10260363.
- O. Gustavsson, **M. Iovino**, J. Styrud, and C. Smith, "Combining Context Awareness and Planning to Learn Behavior Trees from Demonstration," in *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, ©2022 IEEE. Reprinted, with permission., Aug. 2022, pp. 1153–1160. DOI: 10.1109/RO-MAN53752.2022.9900603.

- **M. Iovino**, F. I. Doğan, I. Leite, and C. Smith, “Interactive Disambiguation for Behavior Tree Execution,” in *2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*, ©2022 IEEE. Reprinted, with permission., Nov. 2022, pp. 82–89. DOI: 10.1109/Humanoids53995.2022.10000088.
- **M. Iovino** and C. Smith, “Behavior Trees for Robust Task Level Control in Robotic Applications,” in *Workshop paper at 2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*, 2022.
- J. Styrud, **M. Iovino**, M. Norrlöf, M. Björkman, and C. Smith, “Combining Planning and Learning of Behavior Trees for Robotic Assembly,” in *2022 International Conference on Robotics and Automation (ICRA)*, May 2022, pp. 11 511–11 517. DOI: 10.1109/ICRA46639.2022.9812086.
- **M. Iovino**, J. Styrud, P. Falco, and C. Smith, “Learning Behavior Trees with Genetic Programming in Unpredictable Environments,” in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, ©2021 IEEE. Reprinted, with permission., May 2021, pp. 4591–4597. DOI: 10.1109/ICRA48506.2021.9562088.

Patents

- S. Zudaire, **M. Iovino**, F. Amadio, A. Synodinos, and J. Wang, “Model-free six-dimensional object pose estimation,” en, WO2025228515A1, Nov. 2025.
- J. Styrud and **M. Iovino**, “Combining planning and learning of behavior trees for an industrial robot,” en, SE2150130A1, Feb. 2021.